Data Transformation and Date Modeling

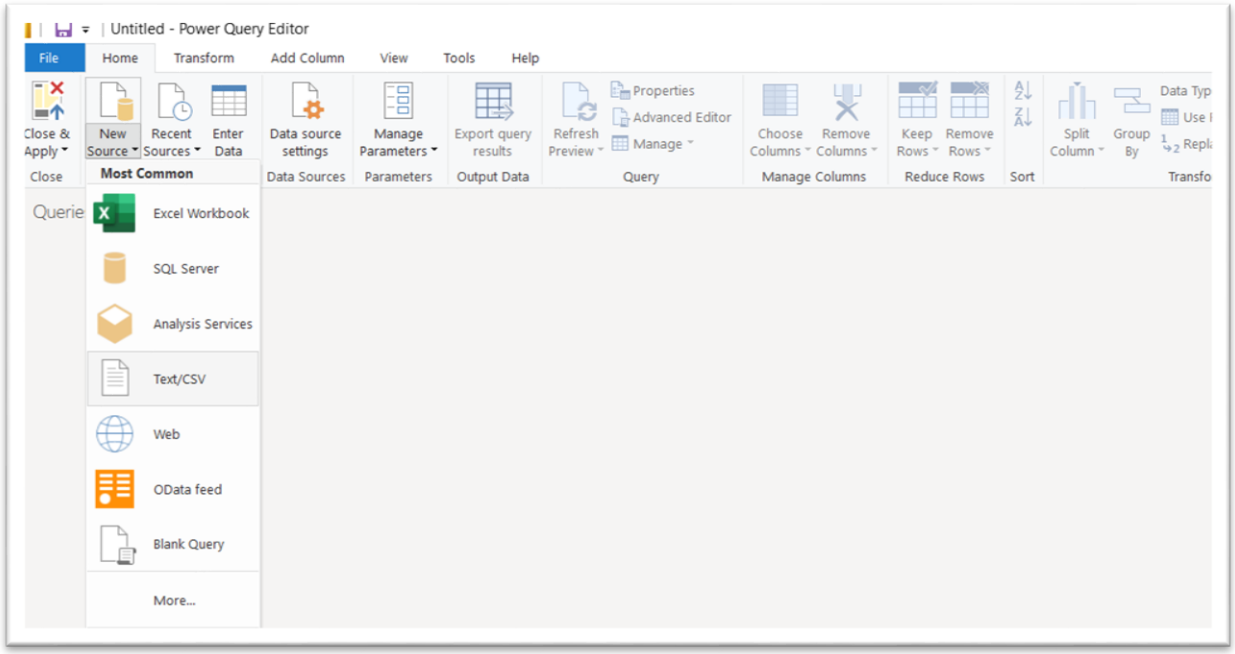
Power BI-Assignment 1

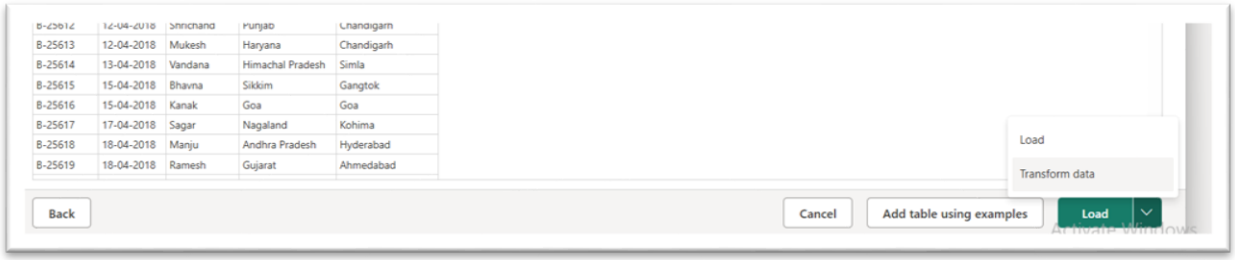
Objective:

To transform, cleanse, and model e-commerce sales data in Power BI, enabling accurate analysis of sales, profitability, customer trends, and target performance through an optimized data model

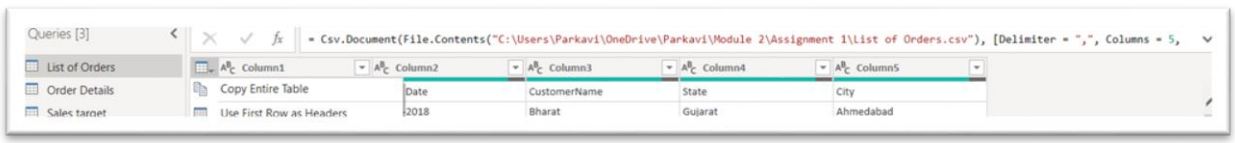
Data Transformation Process:

1. Imported the **List of Orders**, **Order Details**, and **Sales Target** datasets into Power BI

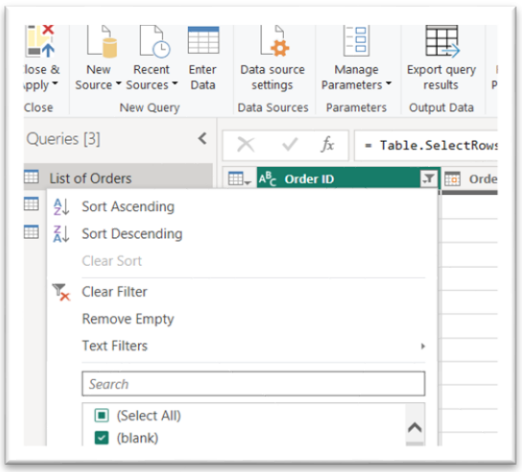


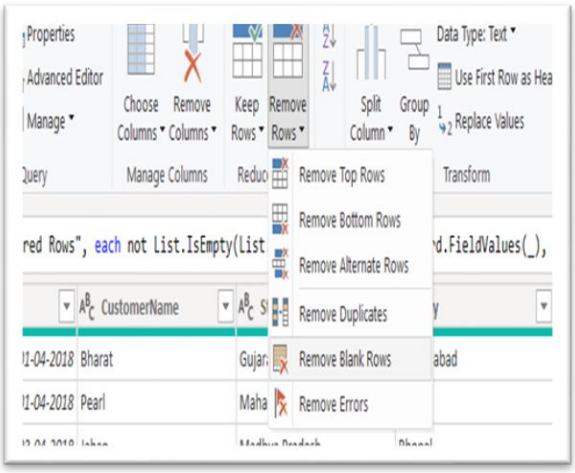


2. Changed the first rows into headers

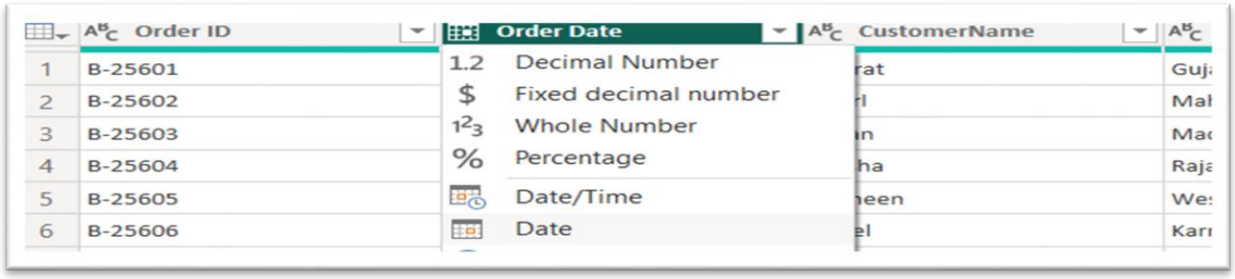


3. Restrict the "List of Orders" table to only the first 500 rows
   (*Cross-checked that there were no blanks in the Order ID by applying a filter*)

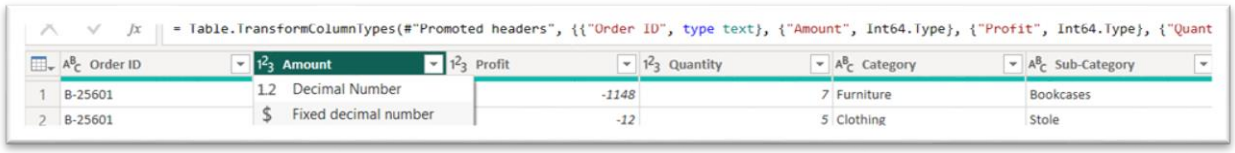




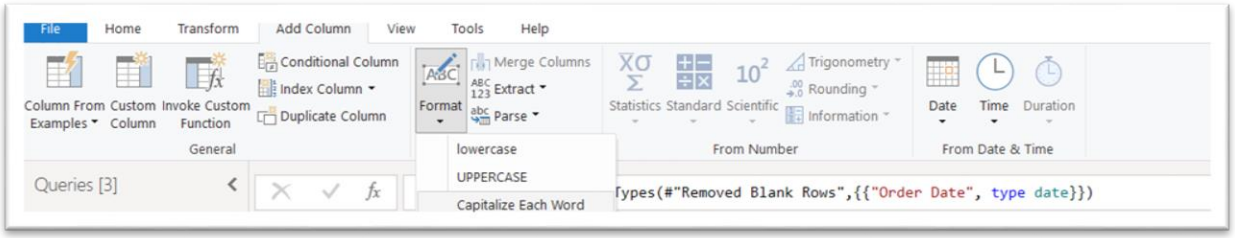
4. Changed the **Order Date** column to the Date data type



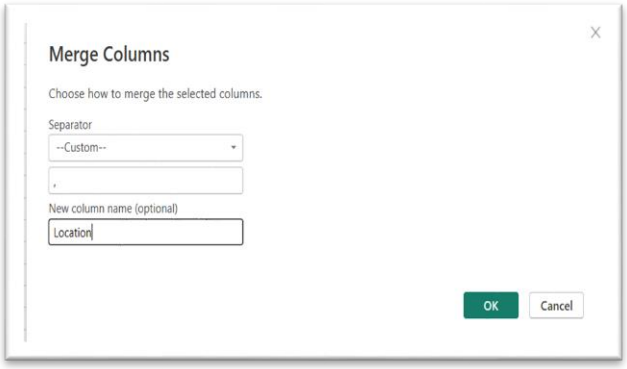
5. Converted the **Amount** and **Target** columns to Fixed Decimal Number data type



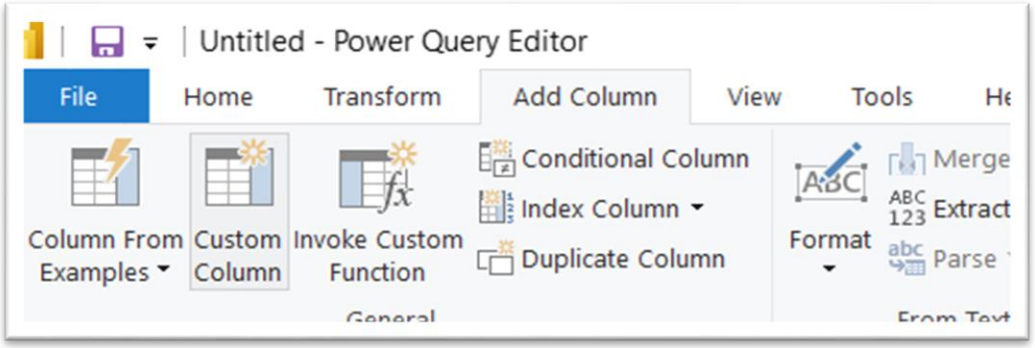
6. Standardized the **Customer Name** column using Proper Case formatting

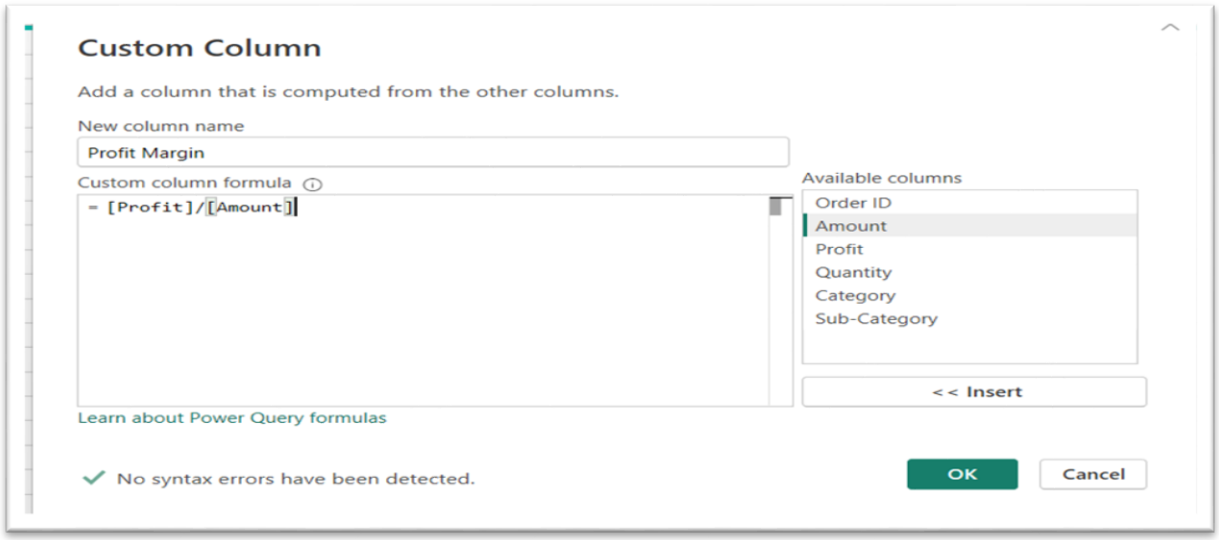


7. Created a **Location** column by combining the City and State fields

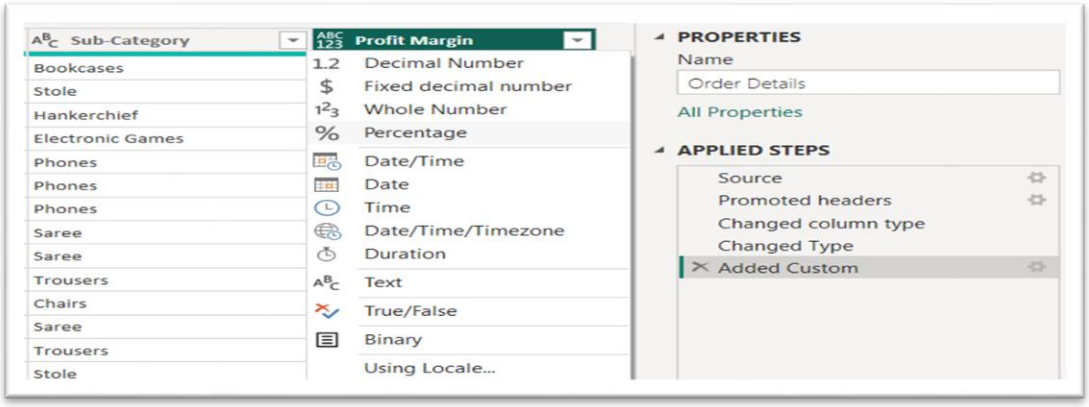


8. Added a **Profit Margin** column to measure profitability

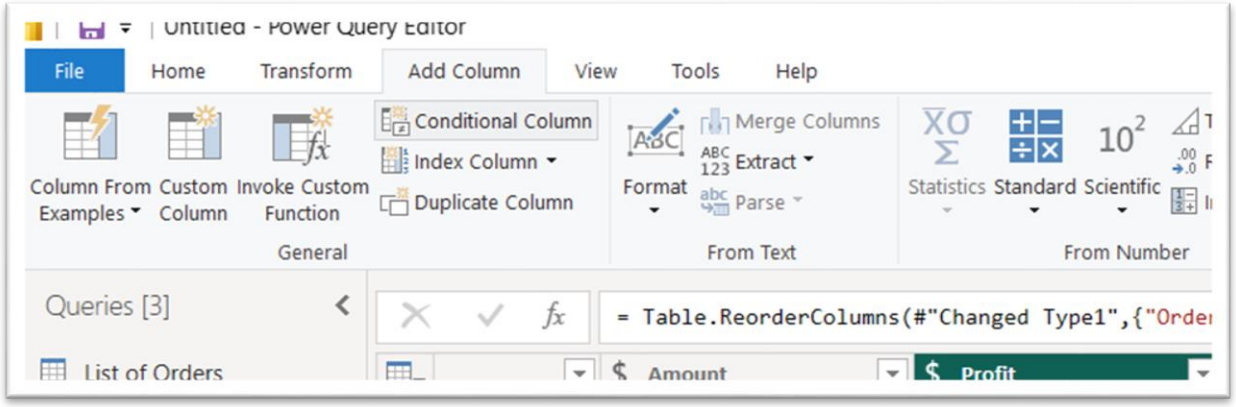


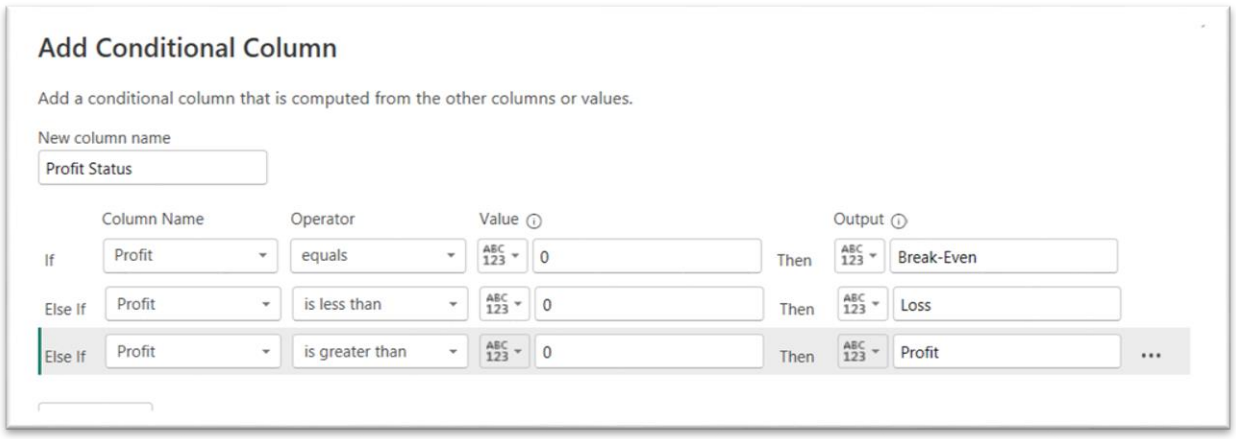


9. Formatted the Profit Margin into Percentage

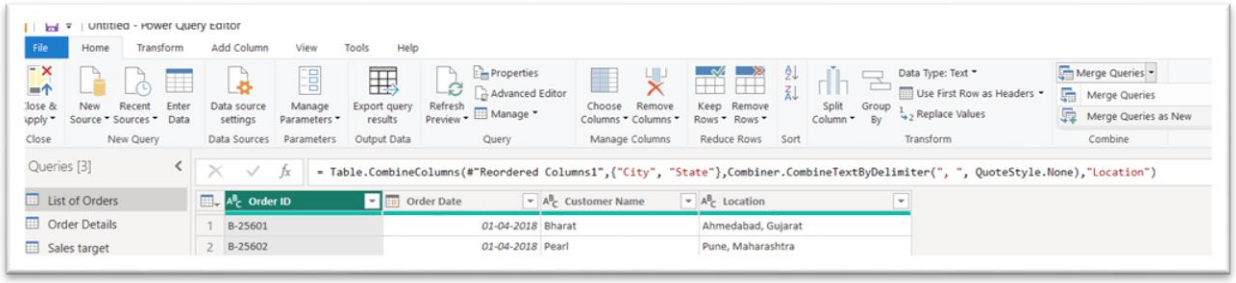


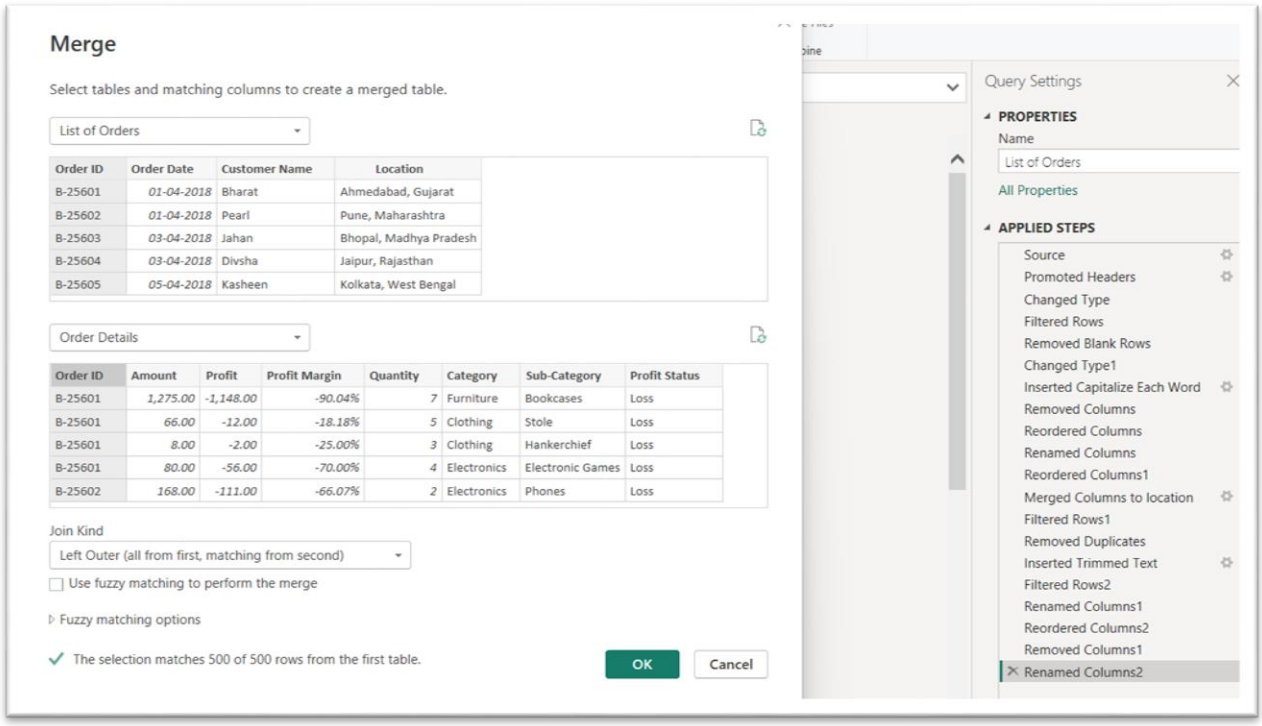
10. Created a **Profit Status** column to categorize records as Profit, Loss, or Break-Even using the Conditional Column option under Add column ribbon

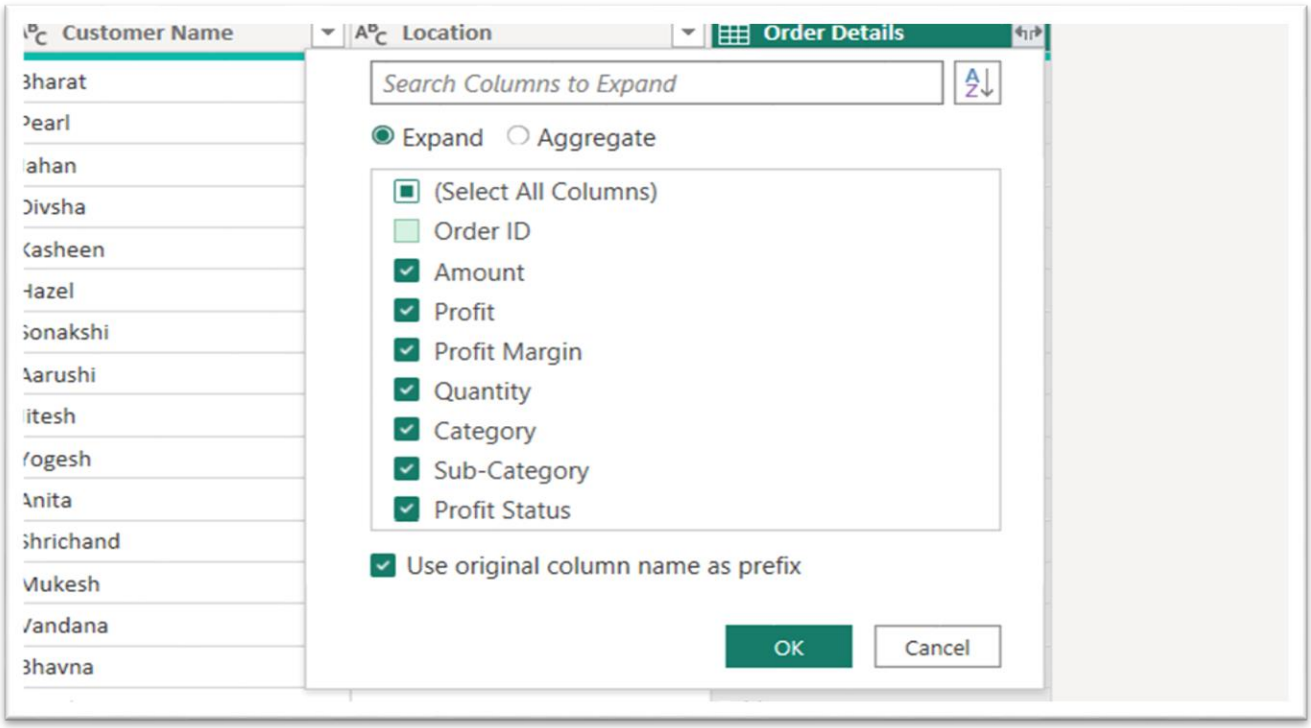




11. Merged the **List of Orders** and **Order Details** tables using the Order ID field to create the **Orders Data** table

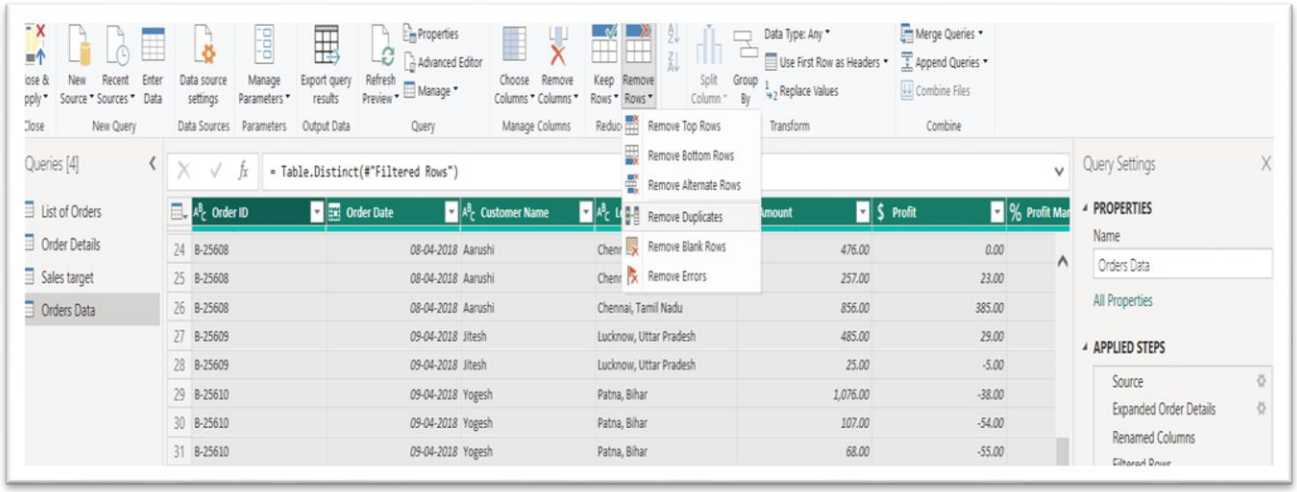

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [3]
List of Orders
Order Details
Sales target
= Table.CombineColumns(#"Reordered Columns1",{"City", "State"},Combiner.CombineTextByDelimiter(", ", QuoteStyle.None),"Location")
Order ID
Order Date
Customer Name
Location
1 B-25601 01-04-2018 Bharat Ahmedabad, Gujarat
2 B-25602 01-04-2018 Pearl Pune, Maharashtra


Merge
Select tables and matching columns to create a merged table.
List of Orders
Order ID | Order Date | Customer Name | Location
B-25601 | 01-04-2018 | Bharat | Ahmedabad, Gujarat
B-25602 | 01-04-2018 | Pearl | Pune, Maharashtra
B-25603 | 03-04-2018 | Jahan | Bhopal, Madhya Pradesh
B-25604 | 03-04-2018 | Divsha | Jaipur, Rajasthan
B-25605 | 05-04-2018 | Kasheen | Kolkata, West Bengal
Order Details
Order ID | Amount | Profit | Profit Margin | Quantity | Category | Sub-Category | Profit Status
B-25601 | 1,275.00 | -1,148.00 | -90.04% | 7 | Furniture | Bookcases | Loss
B-25601 | 66.00 | -12.00 | -18.18% | 5 | Clothing | Stole | Loss
B-25601 | 8.00 | -2.00 | -25.00% | 3 | Clothing | Hankerchief | Loss
B-25601 | 80.00 | -56.00 | -70.00% | 4 | Electronics | Electronic Games | Loss
B-25602 | 168.00 | -111.00 | -66.07% | 2 | Electronics | Phones | Loss
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 500 of 500 rows from the first table.
OK
Cancel
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Filtered Rows
Removed Blank Rows
Changed Type1
Inserted Capitalize Each Word
Removed Columns
Reordered Columns
Renamed Columns
Reordered Columns1
Merged Columns to location
Filtered Rows1
Removed Duplicates
Inserted Trimmed Text
Filtered Rows2
Renamed Columns1
Reordered Columns2
Removed Columns1
Renamed Columns2

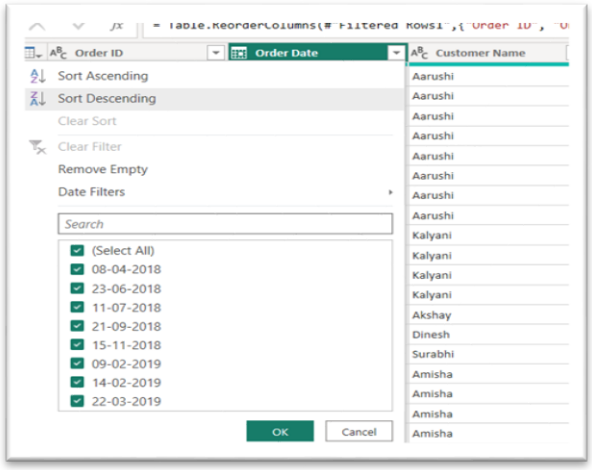

Customer Name
Location
Order Details
Search Columns to Expand
Expand
Aggregate
(Select All Columns)
Order ID
Amount
Profit
Profit Margin
Quantity
Category
Sub-Category
Profit Status
Use original column name as prefix
OK
Cancel

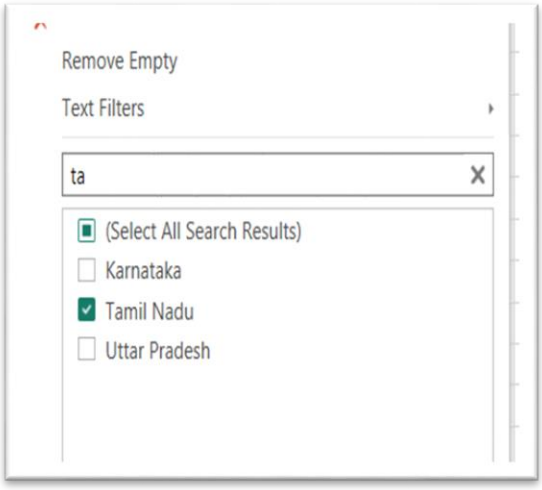
12. Reviewed the dataset for missing values and duplicate records and applied appropriate data quality measures

- Selected all columns and removed exact duplicates. To validate the result, used ***Transform →Statistics → Count Value***s to determine the row count before deduplication
- After removing duplicates, the row count remained 1,500, confirming that no duplicate records existed in the table

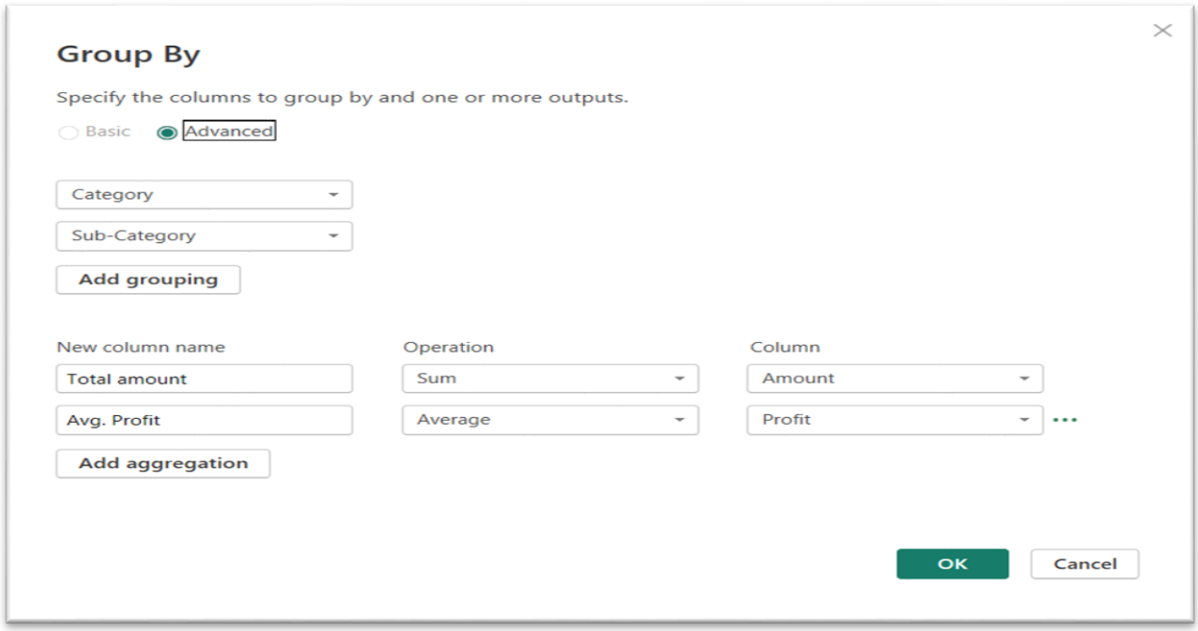


13. Applied sorting and filtering techniques to analyse data by date, state, and category

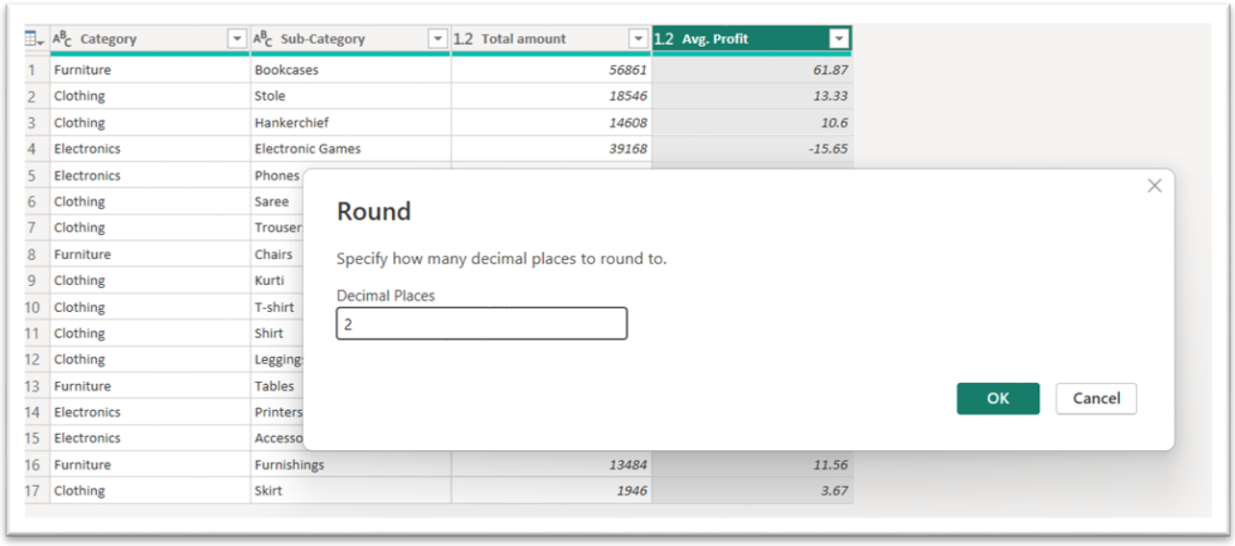




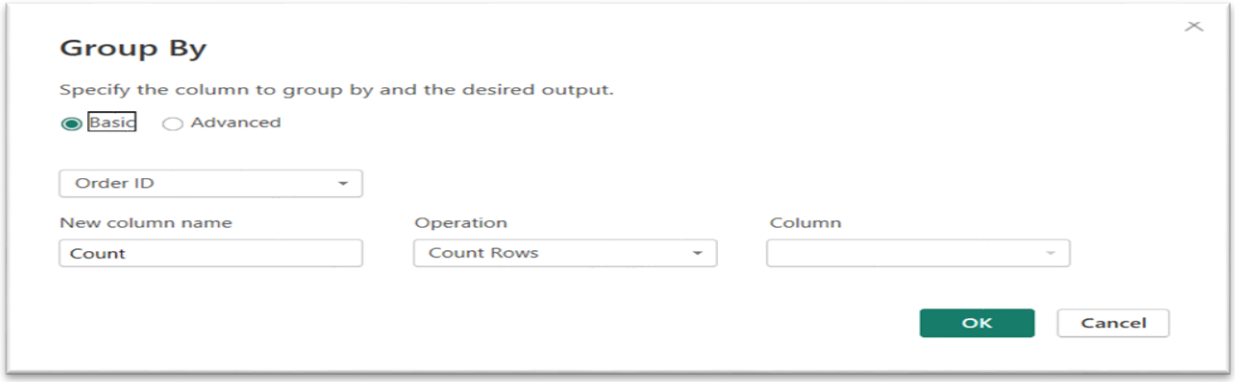
14. Performed grouping and aggregation operations to create summary tables such as Order Count, Order Summary, and Monthly Targets



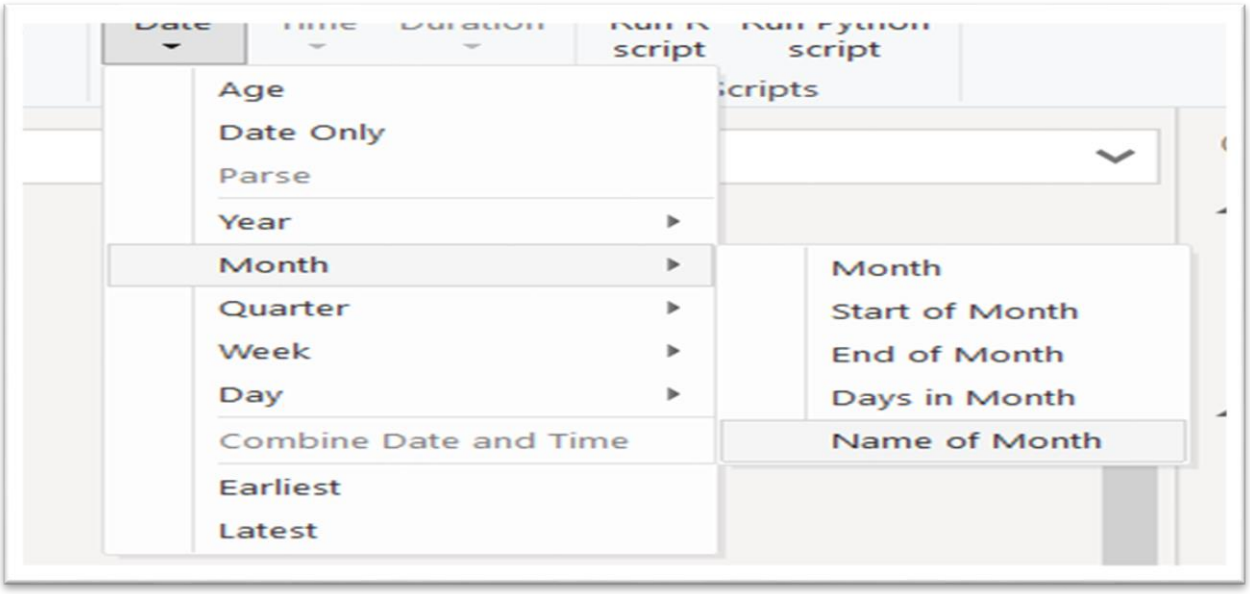
15. Rounded off the Avg. Profit by 2 decimal places



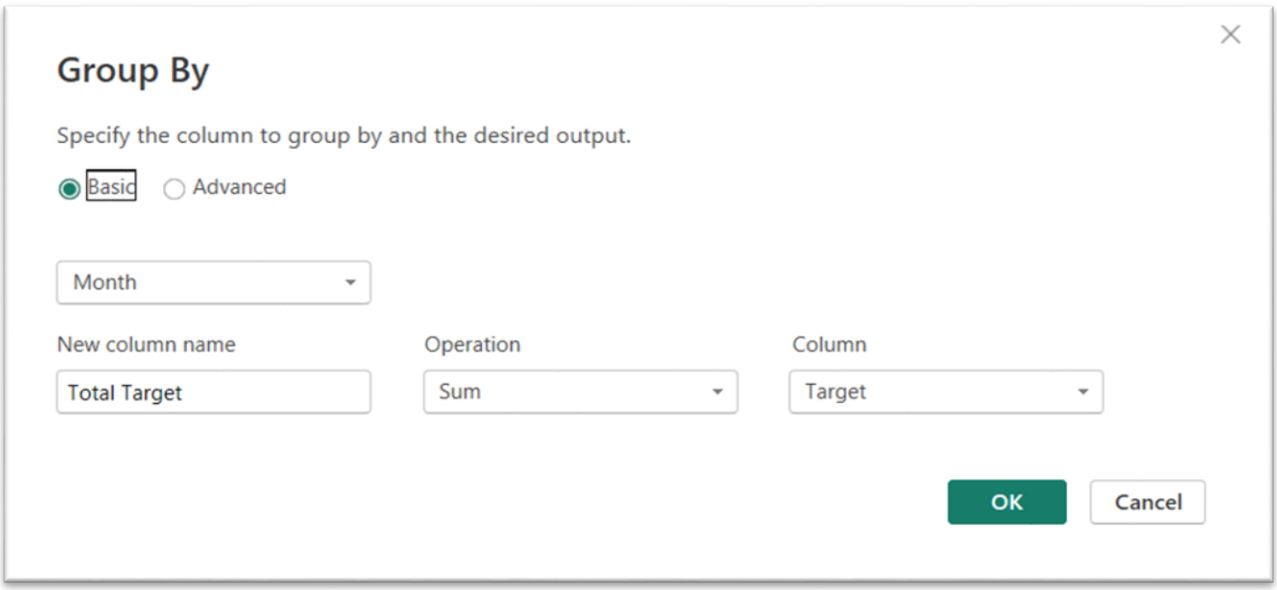
16. Calculated the Order count by grouping Order ID



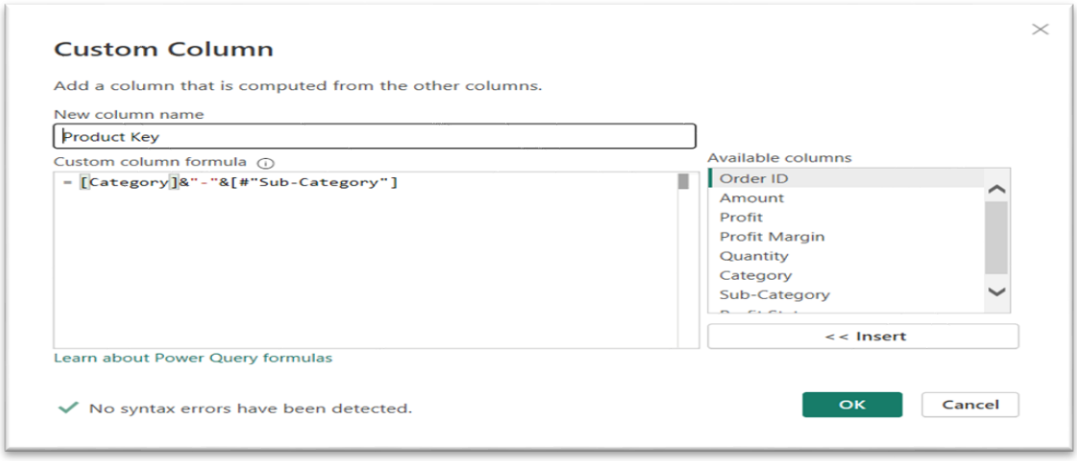
17. Extracted the month name from the *Month of Order Date* column in the Sales Target table using **Tranform-> Date->month-> name of month**



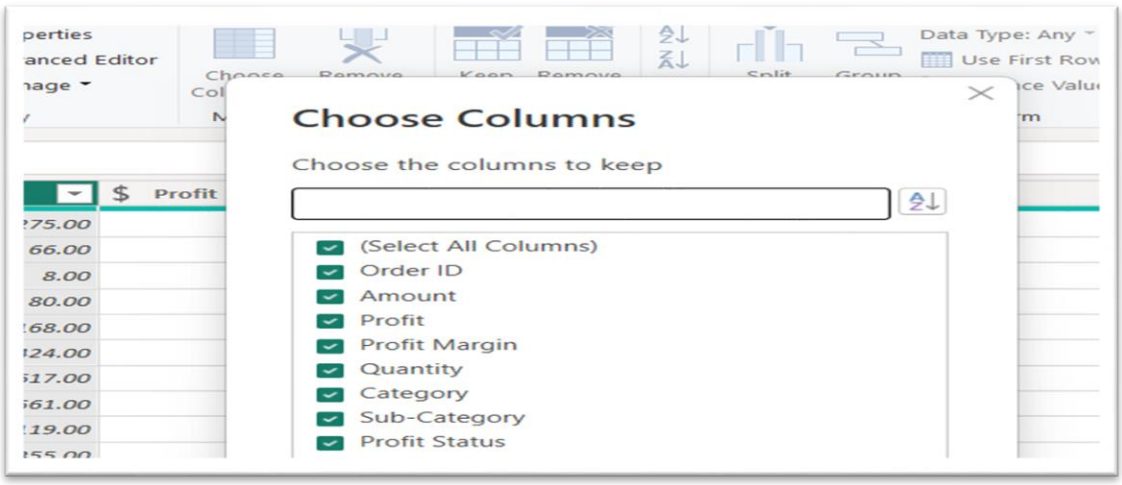
18. Duplicated the table and grouped the data by month to aggregate target values using the **Group By** feature in the **Transform** tab , created **monthly target table**



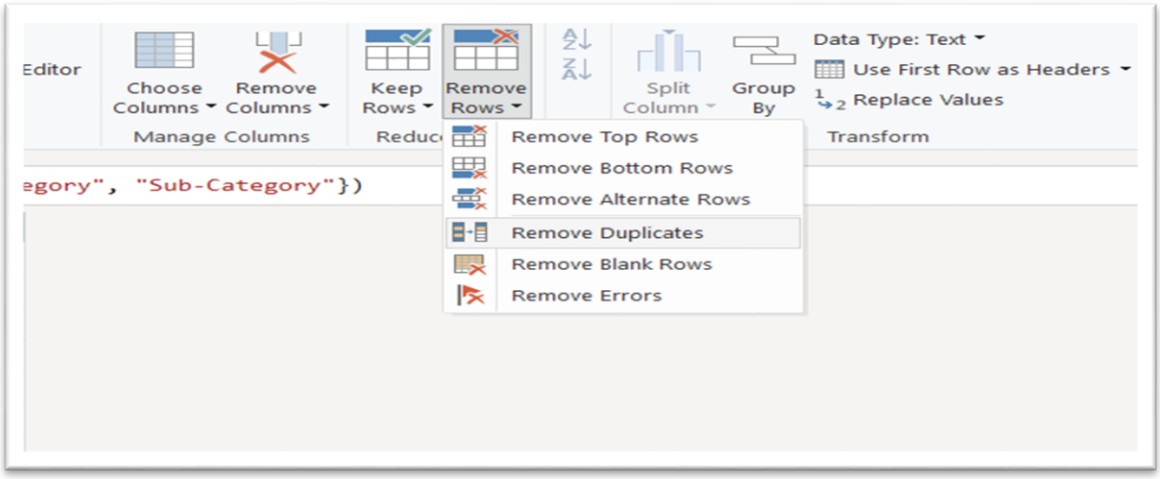
19. Created the Product key for both Order Details and Duplicated the Order details table



20. Renamed the duplicated Order Details as **Dim Product** table and selected the Choose columns to choose columns to keep



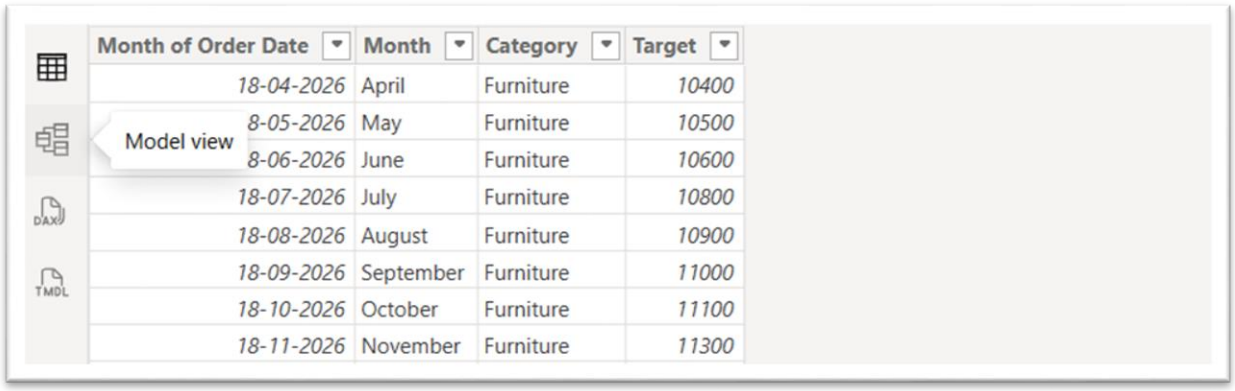
21. Removed the Duplicate Combinations from the entire using Remove Duplicates



22. Finally, selected **Close & Apply** to import the transformed data into Power BI

Data Modelling Process:

1. Moved to Model view in the Power BI

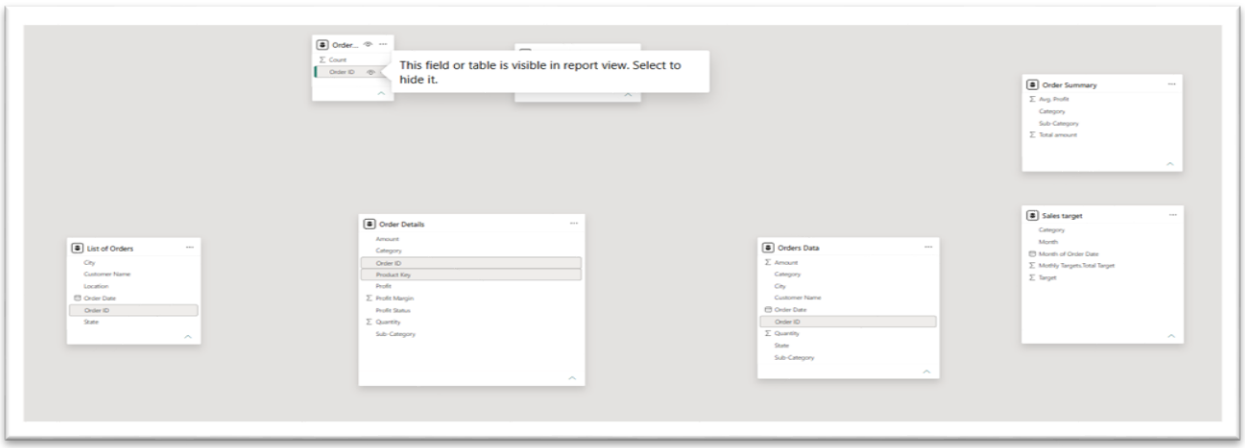


Month of Order Date	Month	Category	Target
18-04-2026	April	Furniture	10400
8-05-2026	May	Furniture	10500
8-06-2026	June	Furniture	10600
18-07-2026	July	Furniture	10800
18-08-2026	August	Furniture	10900
18-09-2026	September	Furniture	11000
18-10-2026	October	Furniture	11100
18-11-2026	November	Furniture	11300

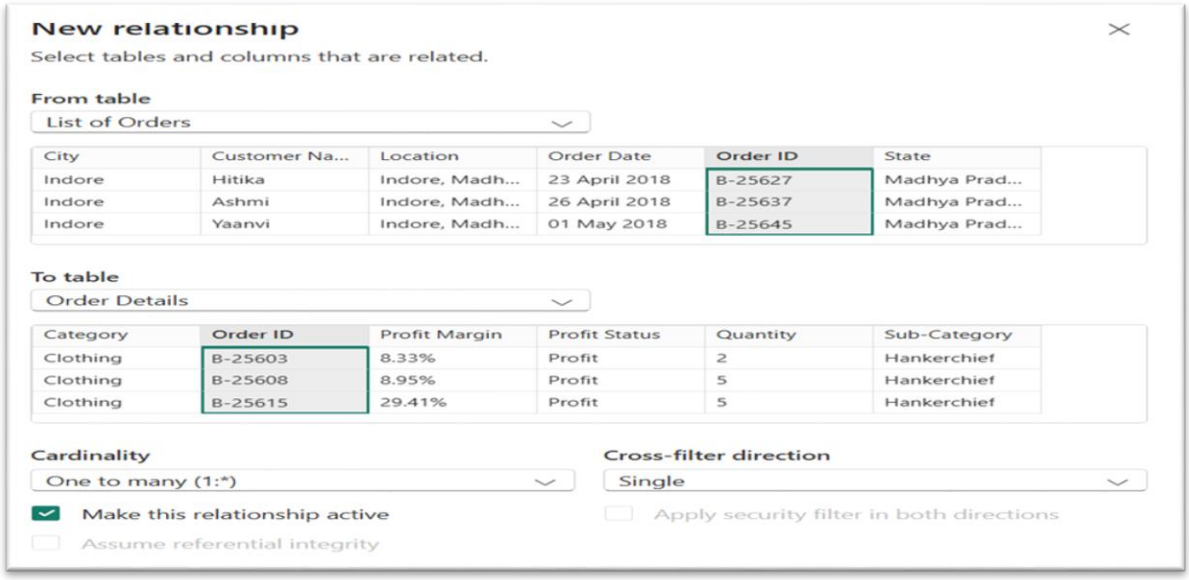
Default view of the Data model by Auto Detect option



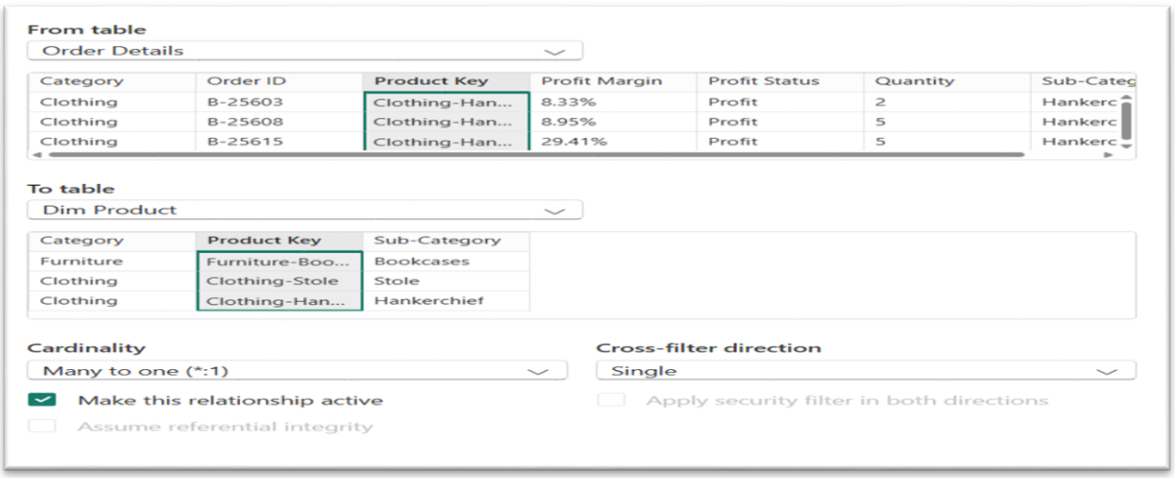
2. Designated Order Details as the primary fact table containing transactional sales data



3. Established a one-to-many relationship between **List of Orders** and **Order Details** using the Order ID field



4. Established a one-to-many relationship between **Dim Product** and **Order Details**



5. Maintained **Order Summary** as a disconnected table for reporting purposes without participating in relationships
   Reason for not Connecting the Orders data
   - **Orders Data** is already a merged table containing Order ID, Customer Name, City, State, Category, Quantity, etc
   - Because both tables contain multiple records for the same **Order ID**, creating a relationship between them will likely result in a **Many-to-Many** relationship, which is not recommended

6. Ensured that the data model followed a simplified star-schema approach with a single fact table and supporting dimension tables

7. Configured relationship directions and cardinality to support accurate filtering and reporting

Final Data Model for E-commerce Data:

